

# DIWALI SALES ANALYSIS REPORT

## 1. Introduction

Diwali is one of the biggest festivals in India and significantly impacts consumer purchasing behavior.

Businesses experience high sales during this period due to discounts, festive offers, and increased customer demand. This project focuses on analyzing Diwali sales data to understand customer behavior and identify important sales trends.

The analysis is performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to extract meaningful insights from the dataset.

## 2. Objectives of the Project

The main objectives of this project are:

- To analyze customer purchasing behavior during Diwali.
- To identify the top-performing states and customer groups.
- To study the impact of age, gender, marital status, and occupation on sales.
- To determine the most purchased product categories.
- To help businesses improve marketing and sales strategies.

## 3. Tools and Technologies Used

The following technologies and tools were used in this project:

- **Python** – Programming language used for analysis.
- **Pandas** – For data cleaning and manipulation.
- **NumPy** – For numerical operations.
- **Matplotlib** – For data visualization.
- **Seaborn** – For statistical graphs and charts.
- **Jupyter Notebook** – For executing analysis and visualizing results.

## 4. Dataset Description

The dataset contains information about customer purchases during Diwali sales. It includes details such as:

- Customer ID
- Gender
- Age Group
- Marital Status
- State
- Occupation
- Product Category
- Orders
- Amount Spent

The dataset was cleaned before analysis by removing missing values and unnecessary columns.

## 5. Data Cleaning and Preprocessing

Data preprocessing is an important step in analysis. The following steps were performed:

- Removed unnecessary columns.
- Checked and handled missing values.
- Converted data types where necessary.
- Removed null values to improve accuracy.
- Organized data for better visualization.

## 6. Exploratory Data Analysis (EDA)

### 6.1 Gender-wise Analysis

The analysis showed that **female customers contributed more to sales than male customers**. Women made higher purchases during the Diwali season.

### 6.2 Age Group Analysis

Customers in the **26–35 age group** contributed the highest number of purchases, making them the major target audience.

### 6.3 State-wise Analysis

States such as **Uttar Pradesh, Maharashtra, and Karnataka** generated the highest sales and order counts.

### 6.4 Marital Status Analysis

**Married customers**, especially women, spent more during Diwali shopping compared to unmarried customers.

### 6.5 Occupation Analysis

Customers working in **IT, Healthcare, and Aviation sectors** contributed significantly to purchases.

### 6.6 Product Category Analysis

The most purchased products belonged to categories such as:

- Food
- Clothing & Apparel
- Electronics

These categories showed high customer demand during the festival season.

## 7. Findings and Insights

The major findings of the project are:

- Female customers purchase more products than male customers.
- Married women aged **26–35 years** are the most active buyers.
- Customers from **Uttar Pradesh, Maharashtra, and Karnataka** contribute heavily to sales.
- IT and Healthcare professionals spend more on festive shopping.
- Food, clothing, and electronics products are highly popular.

## 8. Conclusion

The Diwali Sales Analysis project helps in understanding customer purchasing behavior during festive seasons. The analysis shows that married women between the age group of 26–35 years are the most valuable customers. Businesses can improve sales by targeting these customer groups through personalized marketing campaigns and festival offers.

This analysis helps companies make better business decisions, increase customer satisfaction, and improve overall revenue during festive periods.

## 9. References

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Jupyter Notebook